

Assignment 1: perform CPU scheduling accordingly

Calculate the average waiting time and turnaround time using priority algorithm (Priority 1 being the highest) for the given scenario:

PID	Burst Time	Arrival Time	Priority
A	3	0	3
B	2	2	3
C	4	3	2
D	2	3	1

Assignment 2: perform CPU scheduling accordingly

Why do we need to schedule process? Find the average waiting time and average turnaround time for the following set of processes using FCFS, SJF, RR (Quantum = 3) and shortest remaining time next.

Process	CPU burst time	Arrival time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

Assignment 3: perform page replacement algorithm

How do you recognize critical section? Why do we need to synchronise it? Consider the request for the page references 7,0,1,2,0,3,0,4,2,3,0,3,2. Find the number of page fault for FIFO and LRU with 4 page frames.

Assignment 4: Find the seek time using SCAN, C-SCAN, Look and C-Look disk scheduling algorithms for processing the following request queue:

35, 70, 45, 15, 65, 20, 80, 90, 75, 130.

Suppose the disk has tracks numbered from 0 to 150 and assume the disk arm to be at 30 and moving outward.

Assignment 5: solve the given problem?

How unsafe state differs from deadlocked state? Consider follows initial state and identify wheather requested is and granted or denies for the given cases.

Process	Has	Max
A	2	6
B	1	5
C	2	3
D	3	8

Free = 2

- What will happen if process D request 1 resource?
- What will happen if process A request 1 resource?